1. Implement a STACK ADT with all its operations.

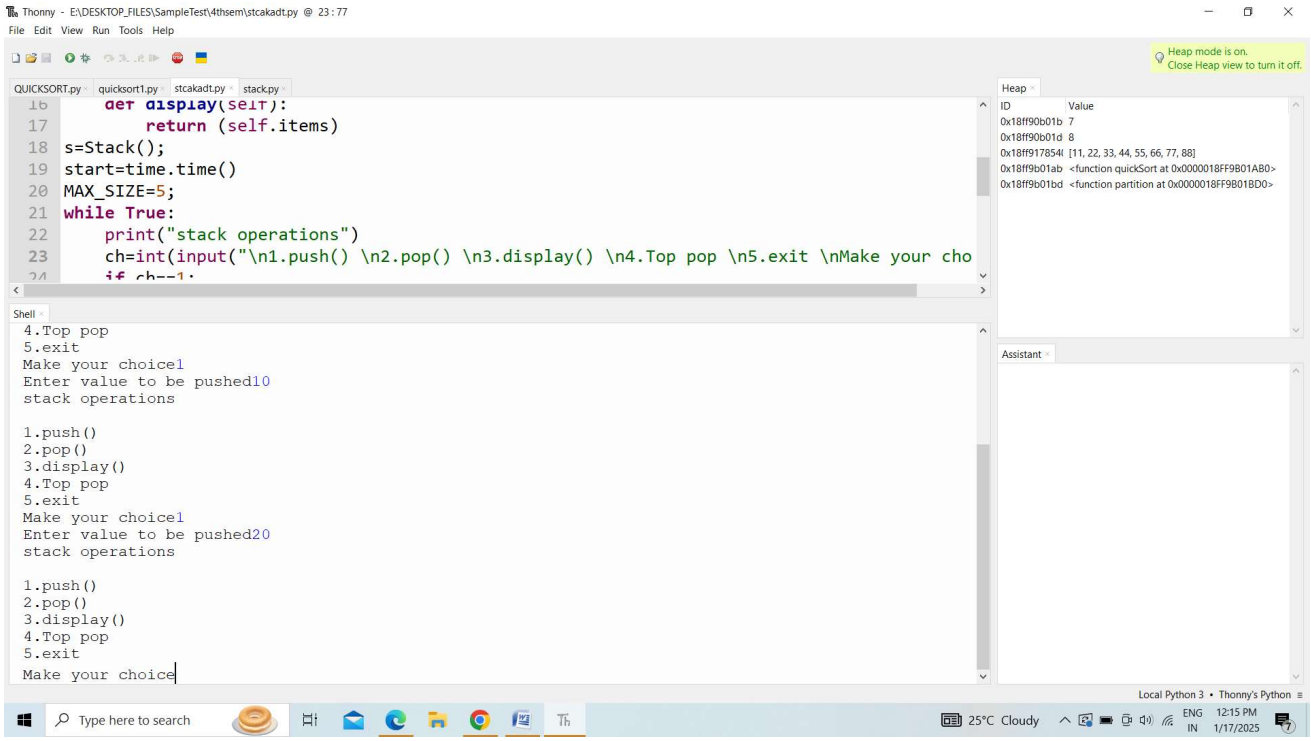
Algorithm:

```
Algorithm:
Class Stack
  items=[]
  def init(self)
      self.items=[]
  def isempty(self)
      returns self.items=[]
  def push(self,item)
      self.items. append(item)
  def pop(self):
      return self.items.pop()
  def peek(self):
      return self.items[len(self.items) - 1]
  def size(self):
      return len(self.items)
  def display(self):
      return (self.items)
```

```
import time
class Stack:
   items=[];
   def init(self):
      self.items = []
   def isEmpty(self):
      return self.items == []
   def push(self, item):
      self.items.append(item)
   def pop(self):
      return self.items.pop()
   def peek(self):
      return self.items[len(self.items) - 1]
   def size(self):
      return len(self.items)
   def display(self):
      return (self.items)
s=Stack();
start=time.time()
MAX_SIZE=5;
```

```
while True:
   print("stack operations")
   ch=int(input("\n1.push()  \n2.pop()  \n3.display()     \n4.Top pop \n5.exit
        \nMake your choice"))
   if ch==1:
      if s.size()== MAX_SIZE:
         print("Stack Overflown")
      else:
         val=input("Enter value to be pushed")
         s.push(val)
    elif ch ==2:
       if s.isEmpty():
         print("Stack Underflow")
      else:
         print("popped item is",s.pop());
   elif ch==3:
      if s.isEmpty():
         print("Stack IS EMPTY")
      else:
         print("Stack Items are");
         print(s.display())
   elif ch==4:
         print("the top element is(peek)",s.peek());
   else:
      break;
end = time.time()
print("Run time of the program is",end-start)
```

Output:


Thonny - E:\DESKTOP_FILES\SampleTest\4thsem\stcakadt.py @ 23 : 77
File Edit View Run Tools Help
Heap mode is on.
Close Heap view to turn it off.
QUICKSORT.py
quicksort1.py
stcakadt.py
stack.py
def display(self):
return (self.items)
s=Stack();
start=time.time()
MAX_SIZE=5;
while True:
print("stack operations")
ch=int(input("\n1.push() \n2.pop() \n3.display() \n4.Top pop \n5.exit \nMake your cho
Heap
ID
Value
0x18ff90b01b 7
0x18ff90b01d 8
0x18ff917854 [11, 22, 33, 44, 55, 66, 77, 88]
0x18ff9b01ab <function quickSort at 0x0000018FF9B01AB0>
0x18ff9b01bd <function partition at 0x0000018FF9B01BD0>
Shell
4.Top pop
5.exit
Make your choice1
Enter value to be pushed10
stack operations
1.push()
2.pop()
3.display()
4.Top pop
5.exit
Make your choice1
Enter value to be pushed20
stack operations
1.push()
2.pop()
3.display()
4.Top pop
5.exit
Make your choice
Assistant
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ENG
IN
12:15 PM
1/17/2025

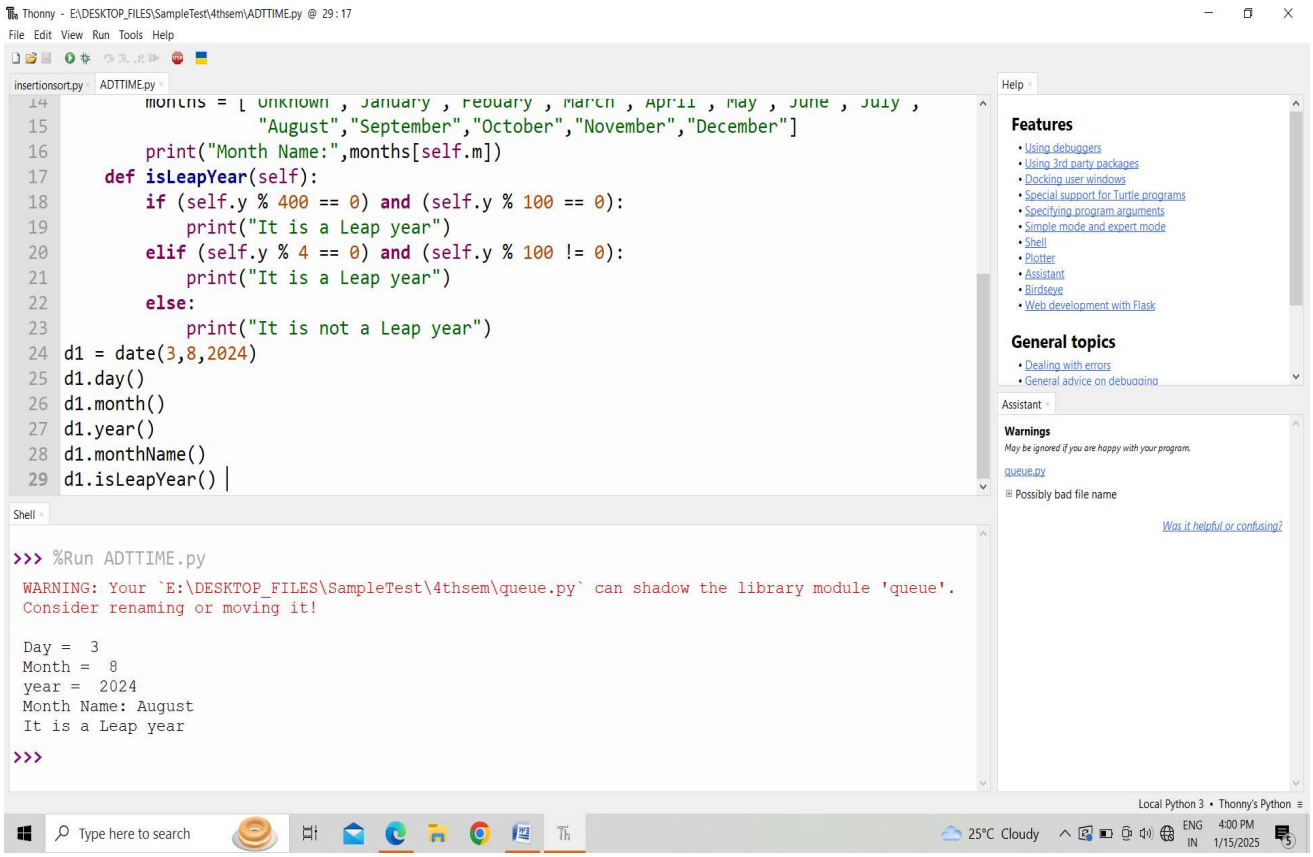
2. Implement a DATE ADT with all its operations.

<u>Algorithm:</u>

```
Algorithm:
class date:
   def  init(self,a,b,c):
        self.d=a
        self.m=b
        self.y=c

   def day(self):
        print("Day = ", self.d)

   def month(self):
        print("Month = ", self.m)

   def year(self):
        print("year = ", self.y)

   def monthName(self):

months=["Unknown","January","Febuary","March","April","May","June
","July", "August", "September", "October", "November", "December"]
      print("Month Name:",months[self.m])

   def isLeapYear(self):
        if (self.y % 400 == 0) and (self.y % 100 == 0):
            print("It is a Leap year")
       elif (self.y % 4 == 0) and (self.y % 100 != 0):
            print("It is a Leap year")
      else:
           print("It is not a Leap year")
```

```
class date:
   def __init__(self,a,b,c):
        self.d=a
        self.m=b
        self.y=c

   def day(self):
        print("Day = ", self.d)
   def month(self):
        print("Month = ", self.m)
   def year(self):
        print("year = ", self.y)
   def monthName(self):

months=["Unknown","January","Febuary","March","April","May","June","
July", "August", "September", "October", "November", "December"]
        print("Month Name:",months[self.m])

   def isLeapYear(self):
         if (self.y % 400 == 0) and (self.y % 100 == 0):
              print("It is a Leap year")
        elif (self.y % 4 == 0) and (self.y % 100 != 0):
              print("It is a Leap year")
        else:
              print("It is not a Leap year")

d1 = date(3, 8, 2024)
d1.day()
d1.month()
d1.year()
d1.monthName()
d1.isLeapYear()
```

Output:



3. Implement Linear Search and compute space and time complexities, Plot graph using asymptomatic notations

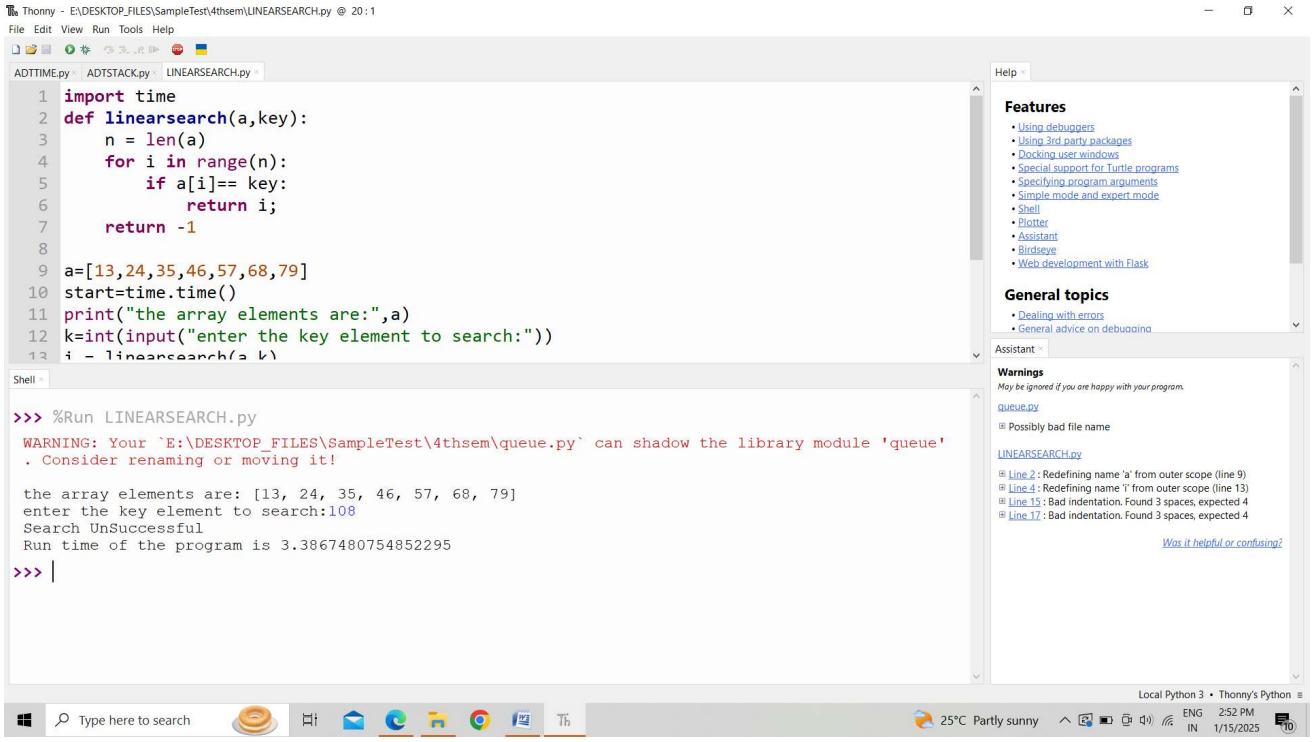
Algorithm:

Algorithm linearsearch(A[0,1,2,---n-1],Key)
Input: // An array A[0,1,2,---n-1],search value is Key
Output: // Returns the index of elements if found,
otherwise returns -1

```
for i in range(n)
    if A[i]== Key
        return i
    return -1
```

```
import time
def linearsearch(a,key):
    n = len(a)
    for i in range(n):
        if a[i]== key:
            return i;
    return -1

a=[13,24,35,46,57,68,79]
start=time.time()
print("the array elements are:",a)
k=int(input("enter the key element to search:"))
i = linearsearch(a,k)
if i==-1:
   print("Search UnSuccessful")
else:
   print("Search Successful key found at location:",i+1)
end = time.time()
print("Run time of the program is",end-start)
```

Output:



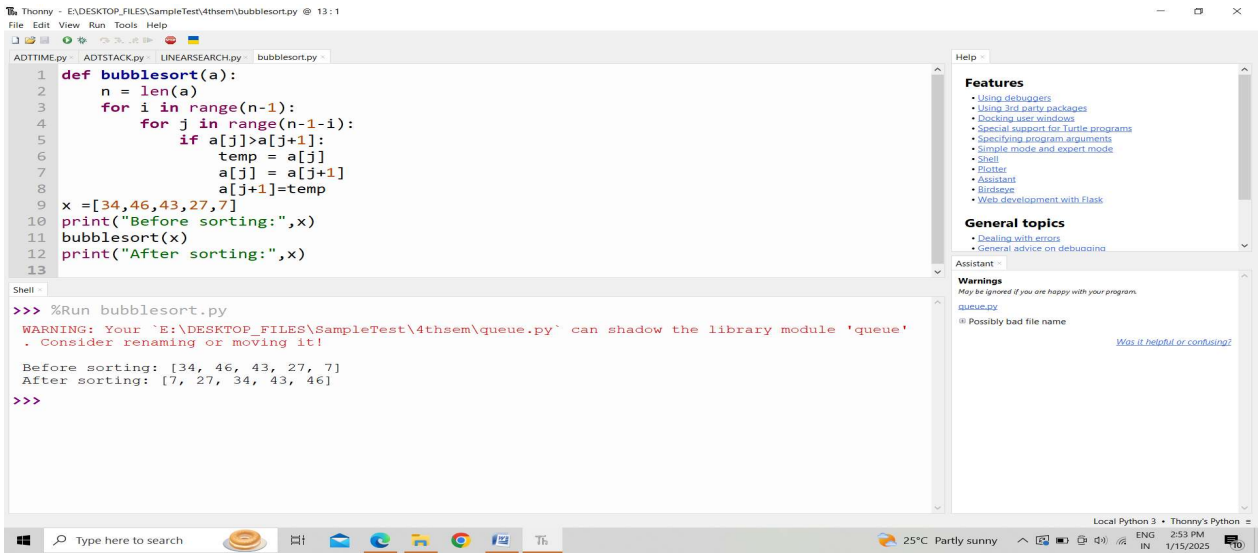
4. Implement Bubble Sort and compute space and time complexities, Plot graph using asymptomatic notation.

Algorithm:

```
Algorithm Bubblesort(A[0,1,2,3, --  n-1],n)
Input:// An array Elements A[0,1,2,3,--n-1],with n elements
Output:// Sorted Array.
    for i in range(n-1)
        for j in range (n-1-i)
            if a[j]>a[j+1]:
                temp = a[j]
                a[j] = a[j+1]
                a[j+1]=temp
```

```
def bubblesort(a):
    n = len(a)
    for i in range(n-1):
        for j in range(n-1-i):
            if a[j]>a[j+1]:
                temp = a[j]
                a[j] = a[j+1]
                a[j+1]=temp
x =[34,46,43,27,7]
print("Before sorting:",x)
bubblesort(x)
print("After sorting:",x)
```

Output:



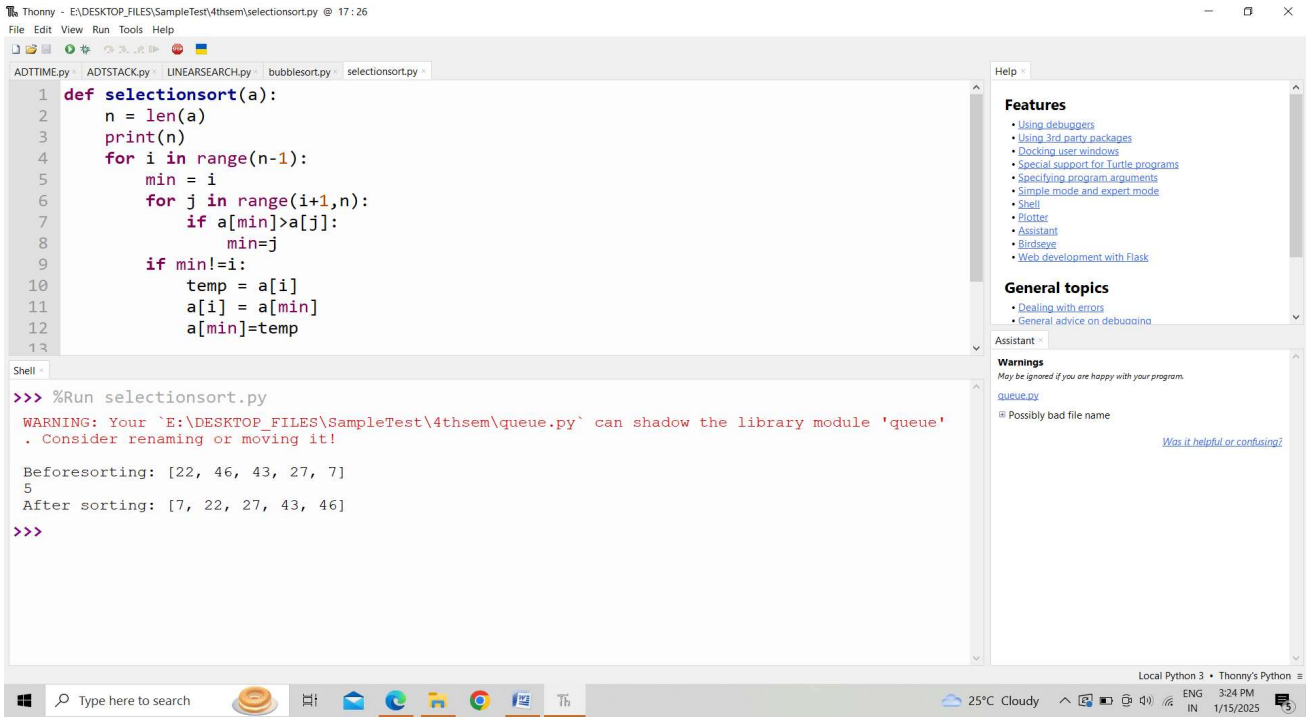
5. Implement Selection Sort and compute space and time complexities, Plot graph using asymptomatic notations

Algorithm:

```
Algorithm Selectionsort(A[0,1,2,3, --  n-1],n)
Input:// An array Elements A[0,1,2,3,--n-1] with n elements
Output:// Sorted Array.
    for i in range(n-1)
        min = i
       for j in range(i+1,n):
           if a[min]>a[j]:
               min=j
       if min!=i:
          temp = a[i]
          a[i] = a[min]
          a[min]=temp
```

```
def selectionsort(a):
    n = len(a)
    print(n)
    for i in range(n-1):
        min = i
        for j in range(i+1,n):
            if a[min]>a[j]:
                min=j
        if min!=i:
           temp = a[i]
           a[i] = a[min]
           a[min]=temp

 x = [22, 46, 43, 27, 7]
print("Before sorting:",x)
selectionsort(x)
print("After sorting:",x)
```

Output:



6. Implement Insertion Sort and compute space and time complexities, Plot graph using asymptomatic notations

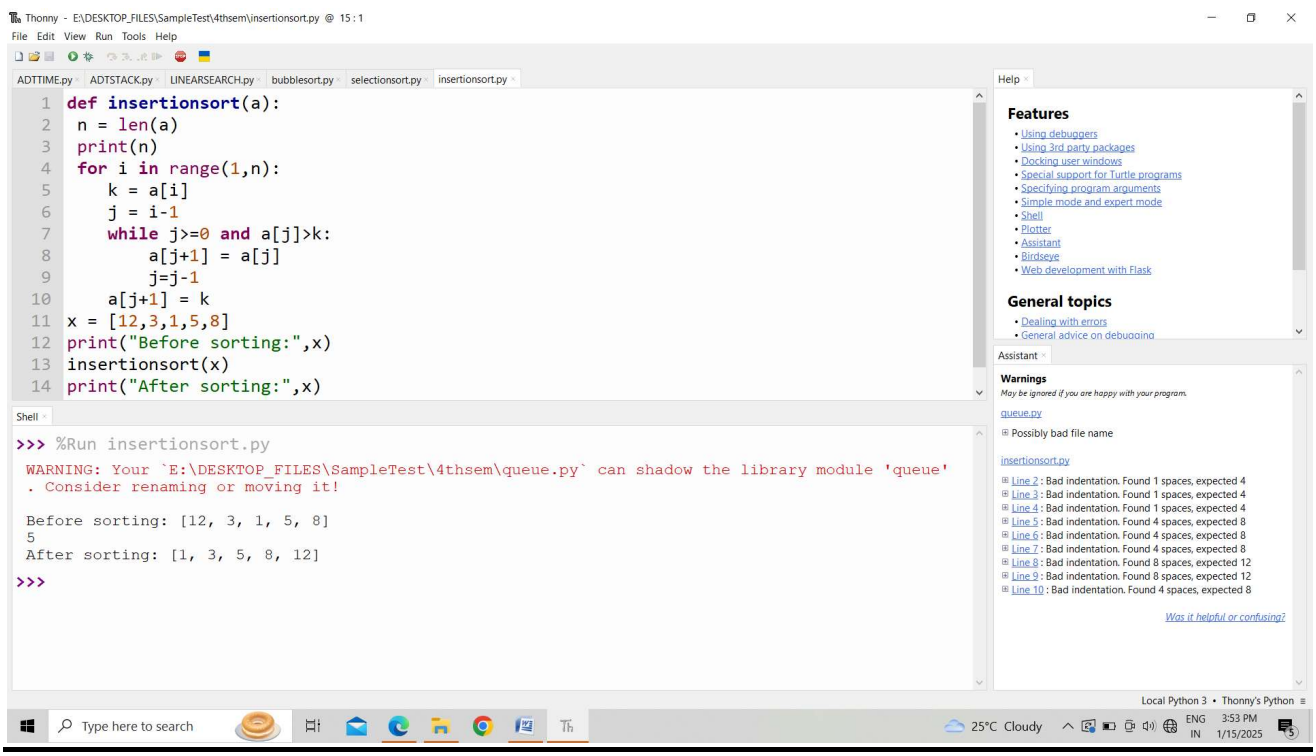
Algorithm:

Algorithm Insertionsort(A[0,1,2,3, -- n-1],n)
Input:// An array Elements A[0,1,2,3,--n-1] with n elements
Output:// Sorted Array.

```
for i in range(1,n):
    k = a[i]
    j = i-1
    while j>=0 and a[j]>k:
        a[j+1] = a[j]
        j=j-1
    a[j+1] = k
```

```
def insertionsort(a):
     n = len(a)
    print(n)
    for i in range(1,n):
          k = a[i]
          j = i-1
         while j>=0 and a[j]>k:
                a[j+1] = a[j]
                 j=j-1
        a[j+1] = k
  x = [12, 3, 1, 5, 8]
  print("Before sorting:",x)
  insertionsort(x)
  print("After sorting:",x)
```

Output:



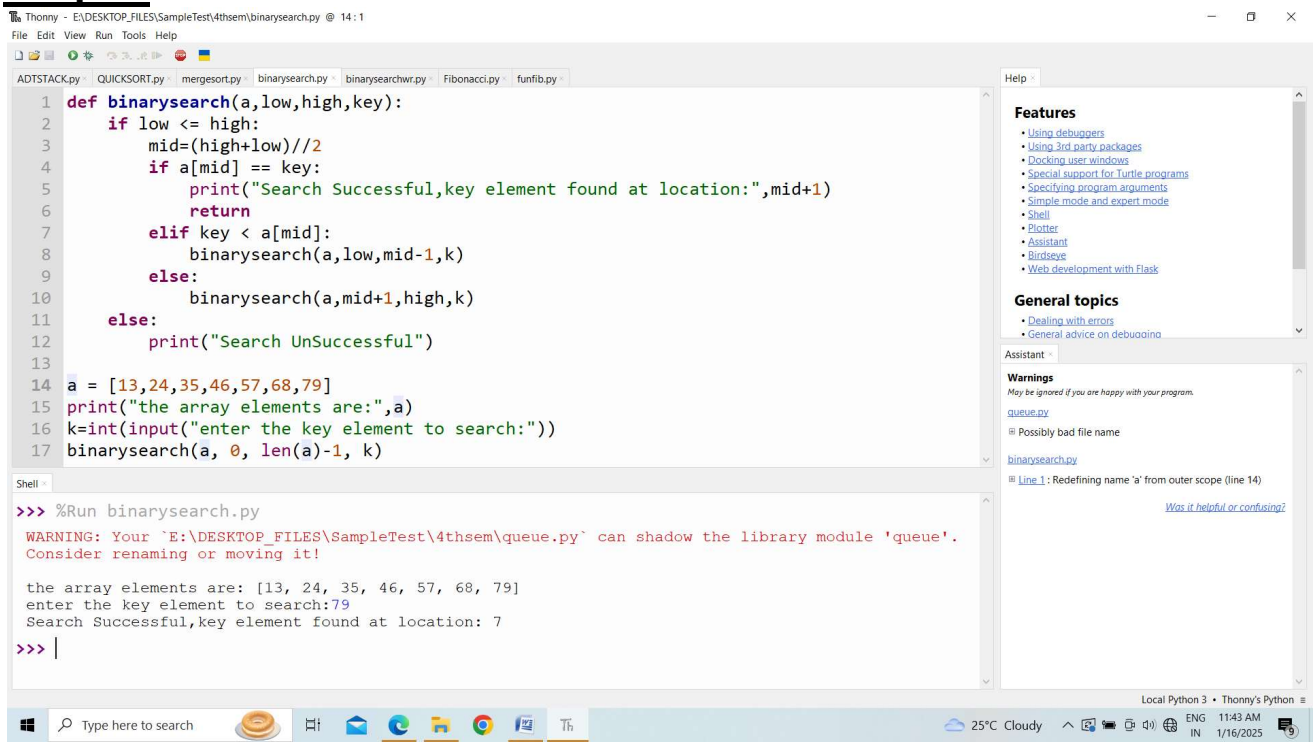
7. Implement Binary Search using Recursion and compute space and Time complexities, plot graph using asymptomatic notations.

Algorithm:

```
Algorithm Binarysearch(A[0,1,2,--n], low, high, key)

Input: //An array Elements A[0,1,2,3,--n] with n elements in
          sorted order and Low and High values along with key
          value as a inputs
Output: // Returns  the index of elements if found,
             Otherwise returns -1
      if  low <= high:
          mid=(high+low)//2
          if a[mid] == key:
               return mid
         elif  key < a[mid]:
               binarysearch(a, low, mid-1, k)
         else:
               binarysearch(a, mid+1, high, k)
      else:
          print("Search UnSuccessful")
```

```
def  binarysearch(a, low, high, key):
      if  low <= high:
          mid=(high+low)//2
          if a[mid] == key:
               print("Search Successful,key element found at location:",mid+1)
               return
          elif  key < a[mid]:
               binarysearch(a,low,mid-1,k)
          else:
               binarysearch(a,mid+1,high,k)
      else:
          print("Search UnSuccessful")

a = [13, 24, 35, 46, 57, 68, 79]
print("the array elements are:",a)
k=int(input("enter the key element to search:"))
binarysearch(a, 0, len(a)-1, k)
```

Output:



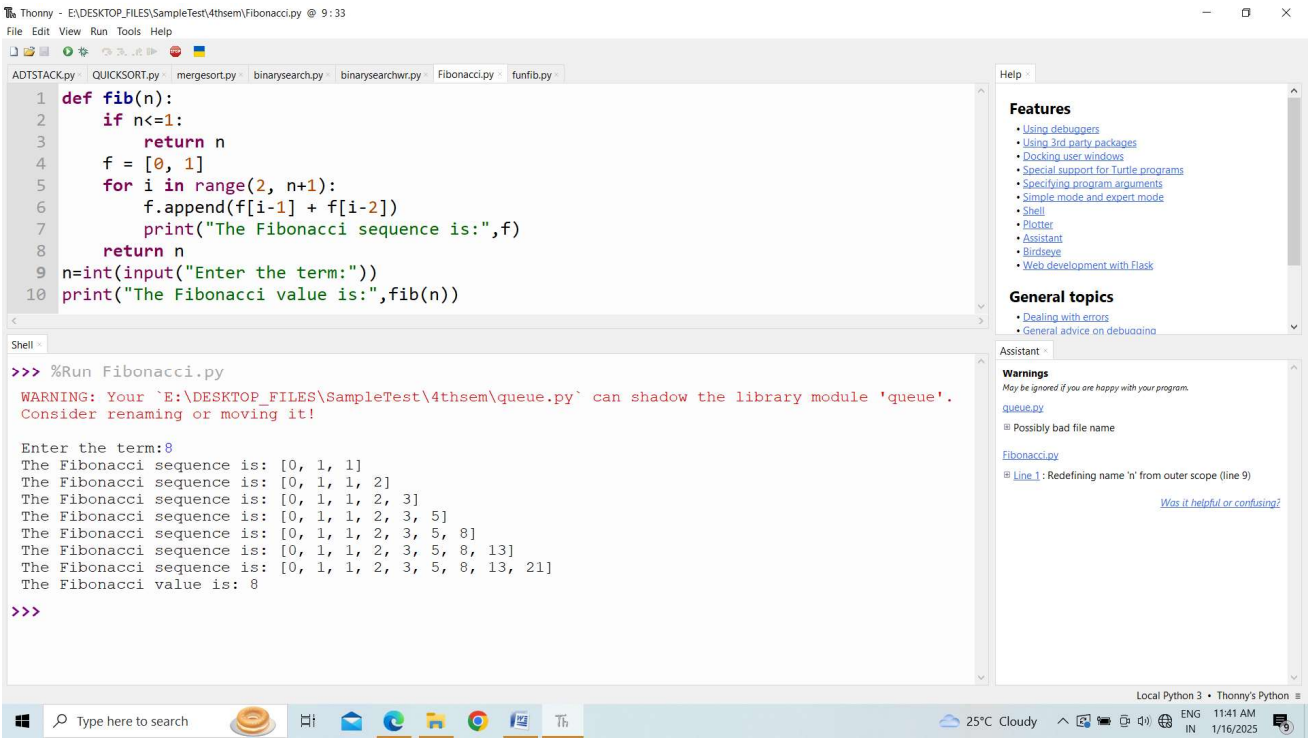
8. Implement Fibonacci sequence with dynamic programming.

Algorithm:

```
Algorithm def fib(n):
Input://Input value is n.
Output:// Fibonacci value:
        if n<=1
          return n
        f=[0,1]
        for i in range(2, n+1)
            f.append(f[i-1] + f[i-2])
            print("The Fibonacci sequence is:",f)
```

```
def fib(n):
    if n<=1:
        return n
   f = [0, 1]
   for i in range(2, n+1):
        f.append(f[i-1] + f[i-2])
        print("The Fibonacci sequence is:",f)
   return n
n=int(input("Enter the term:"))
print("The Fibonacci value is:",fib(n))
```

Output:


Thonny - E:\DESKTOP_FILES\SampleTest\4thsem\Fibonacci.py @ 9 : 33
File Edit View Run Tools Help
ADTSTACK.py QUICKSORT.py mergesort.py binarysearch.py binarysearchwr.py Fibonacci.py funfib.py
def fib(n):
if n<=1:
return n
f = [0, 1]
for i in range(2, n+1):
f.append(f[i-1] + f[i-2])
print("The Fibonacci sequence is:",f)
return n
n=int(input("Enter the term:"))
print("The Fibonacci value is:",fib(n))
Help
Features
Using debuggers
Using 3rd party packages
Docking user windows
Special support for Turtle programs
Specifying program arguments
Simple mode and expert mode
Shell
Plotter
Assistant
Birdseye
Web development with Flask
General topics
Dealing with errors
Shell
>>> %Run Fibonacci.py
WARNING: Your `E:\DESKTOP_FILES\SampleTest\4thsem\queue.py` can shadow the library module 'queue'.
Consider renaming or moving it!
Enter the term:8
The Fibonacci sequence is: [0, 1, 1]
The Fibonacci sequence is: [0, 1, 1, 2]
The Fibonacci sequence is: [0, 1, 1, 2, 3]
The Fibonacci sequence is: [0, 1, 1, 2, 3, 5]
The Fibonacci sequence is: [0, 1, 1, 2, 3, 5, 8]
The Fibonacci sequence is: [0, 1, 1, 2, 3, 5, 8, 13]
The Fibonacci sequence is: [0, 1, 1, 2, 3, 5, 8, 13, 21]
The Fibonacci value is: 8
>>>
Assistant
Warnings
May be ignored if you are happy with your program.
queue.py
Possibly bad file name
Fibonacci.py
Line 1 : Redefining name 'n' from outer scope (line 9)
Was it helpful or confusing?
Local Python 3 • Thonny's Python
Type here to search
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1/16/2025

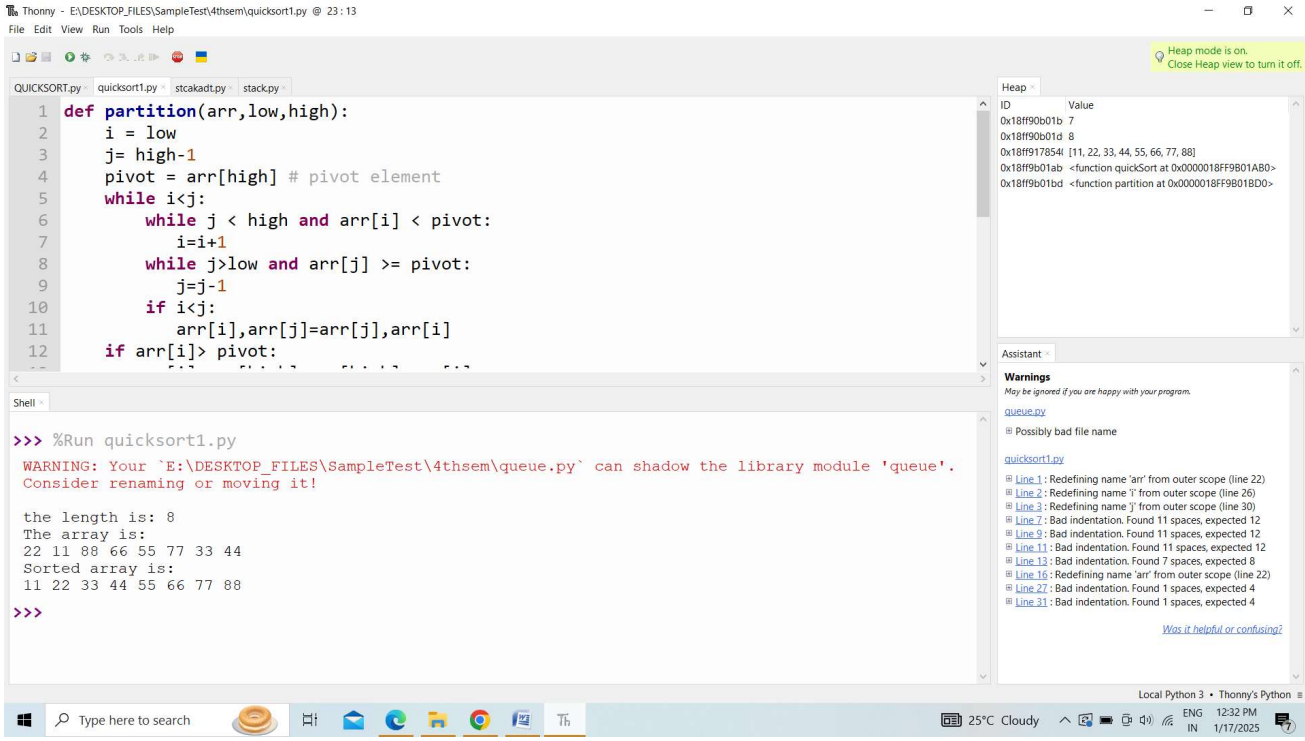
9. Implement quick sorting algorithms compute space and time complexities, plot graph using asymptomatic notations

Algorithm:

```
Algorithm: def partition (arr, low, high):

Input: //An array Elements A[0,1,2,3,--n] with n elements in
          Unsorted order and low and high values as a inputs
Output: // Sorted Array.

   i = low
   j= high-1
   pivot = arr[high] # pivot element
   while i<j:
       while j < high and arr[i] < pivot:
            i=i+1
      while j>low and arr[j] >= pivot:
           j=j-1
      if i<j:
          arr[i],arr[j]=arr[j],arr[i]
  if arr[i]> pivot:
         arr[i],arr[high]=arr[high],arr[i]

 return i

def quickSort(arr,low,high):

   if low < high:
      pi = partition(arr, low, high)
      quickSort(arr,  low,  pi-1)
      quickSort(arr, pi+1, high)
```

```
def partition(arr,low,high):
    i = low
    j= high-1
    pivot = arr[high] # pivot element
    while i < j:
        while j <  high and arr[i] < pivot:
            i=i+1
        while j > low and arr[j] >= pivot:
            j=j-1
        if i<j:
           arr[i],arr[j]=arr[j],arr[i]
    if arr[i] > pivot:
           arr[i],arr[high]=arr[high],arr[i]
   return i

def quickSort(arr, low, high):
    if low < high:
       pi = partition(arr,low,high)
       quickSort(arr, low, pi-1)
       quickSort(arr, pi+1, high)

arr = [22, 11, 88, 66, 55, 77, 33, 44]
n = len(arr)
print("the length is:",n)
print("The array is:")
for i in range(n):
print(arr[i], end=" ")
quickSort(arr,0,n-1)
print("\nSorted array is:")
for j in range(n):
print(arr[j ], end=" ")
```

Output:


Thonny - E:\DESKTOP_FILES\SampleTest\4thsem\quicksort1.py @ 23 : 13
File Edit View Run Tools Help
Heap mode is on.
Close Heap view to turn it off.
QUICKSORT.py quicksort1.py stcakadt.py stack.py
def partition(arr,low,high):
i = low
j= high-1
pivot = arr[high] # pivot element
while i<j:
while j < high and arr[i] < pivot:
i=i+1
while j>low and arr[j] >= pivot:
j=j-1
if i<j:
arr[i],arr[j]=arr[j],arr[i]
if arr[i]> pivot:
Heap
ID Value
0x18ff90b01b 7
0x18ff90b01d 8
0x18ff917854 [11, 22, 33, 44, 55, 66, 77, 88]
0x18ff9b01ab <function quickSort at 0x0000018FF9B01AB0>
0x18ff9b01bd <function partition at 0x0000018FF9B01BD0>
Shell
>>> %Run quicksort1.py
WARNING: Your `E:\DESKTOP_FILES\SampleTest\4thsem\queue.py` can shadow the library module 'queue'.
Consider renaming or moving it!
the length is: 8
The array is:
22 11 88 66 55 77 33 44
Sorted array is:
11 22 33 44 55 66 77 88
>>>
Assistant
Warnings
May be ignored if you are happy with your program.
queue.py
Possibly bad file name
quicksort1.py
Line 1 : Redefining name 'arr' from outer scope (line 22)
Line 2 : Redefining name 'i' from outer scope (line 26)
Line 3 : Redefining name 'j' from outer scope (line 30)
Line 7 : Bad indentation. Found 11 spaces, expected 12
Line 9 : Bad indentation. Found 11 spaces, expected 12
Line 11 : Bad indentation. Found 11 spaces, expected 12
Line 13 : Bad indentation. Found 7 spaces, expected 8
Line 16 : Redefining name 'arr' from outer scope (line 22)
Line 27 : Bad indentation. Found 1 spaces, expected 4
Line 31 : Bad indentation. Found 1 spaces, expected 4
Was it helpful or confusing?
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Type here to search
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ENG IN
12:32 PM
1/17/2025

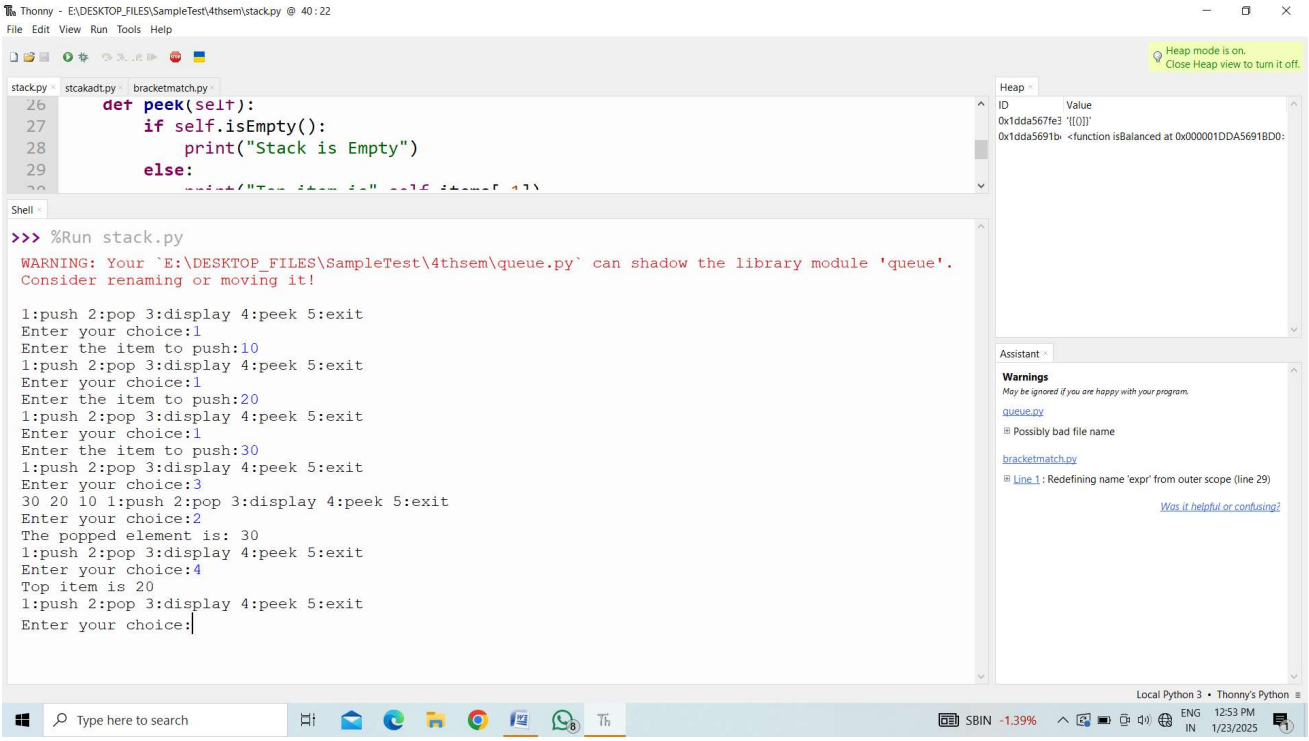
10. Implement Stack Data Structure.

Algorithm:

```
class Stack:
   items=[];
   def init(self):
      self.items = []

   def isEmpty(self):
      return len(self.items)==0

   def push(self,item):
      self.items.append(item)

   def pop(self):
      if self.isEmpty():
         print("Stack is Empty")
      else:
         item=self.items[-1]
         del(self.items[-1])
         print("The popped element is:",item)

     def display(self):
      if self.isEmpty():
         print("Stack is Empty")
      else:
         for i in reversed(self.items):
            print(i,end =' ')
   def peek(self):
      if self.isEmpty():
         print("Stack is Empty")
      else:
         print("Top item is",self.items[-1])
```

```
class Stack:
    items=[];
    def init(self):
        self.items = []

    def isEmpty(self):
        return len(self.items)==0

    def push(self,item):
        self.items.append(item)

    def pop(self):
        if self.isEmpty():
            print("Stack is Empty")
        else:
            item=self.items[-1]
            del(self.items[-1])
            print("The popped element is:",item)

    def display(self):
        if self.isEmpty():
            print("Stack is Empty")
        else:
            for i in reversed(self.items):
                print(i,end =' ')
    def peek(self):
        if self.isEmpty():
            print("Stack is Empty")
        else:
            print("Top item is",self.items[-1])
s = Stack()
while(True):
    print("1:push 2:pop 3:display 4:peek 5:exit")
    choice = int(input("Enter your choice:"))
    if choice == 1:
        val=input("Enter the item to push:")
        s.push(val)
    elif choice == 2:
         s.pop()
    elif choice == 3:
       s.display()
    elif choice == 4:
       s.peek()
    else:
       break
```

Output:


Thonny - E:\DESKTOP_FILES\SampleTest\4thsem\stack.py @ 40 : 22
File Edit View Run Tools Help
stack.py
stcakadt.py
bracketmatch.py
def peek(self):
if self.isEmpty():
print("Stack is Empty")
else:
Heap mode is on.
Close Heap view to turn it off.
Heap
ID
Value
0x1dda567fe3
'[{0}]'
0x1dda5691b
<function isBalanced at 0x000001DDA5691BD0:
Shell
>>> %Run stack.py
WARNING: Your `E:\DESKTOP_FILES\SampleTest\4thsem\queue.py` can shadow the library module 'queue'.
Consider renaming or moving it!
1:push 2:pop 3:display 4:peek 5:exit
Enter your choice:1
Enter the item to push:10
1:push 2:pop 3:display 4:peek 5:exit
Enter your choice:1
Enter the item to push:20
1:push 2:pop 3:display 4:peek 5:exit
Enter your choice:1
Enter the item to push:30
1:push 2:pop 3:display 4:peek 5:exit
Enter your choice:3
30 20 10 1:push 2:pop 3:display 4:peek 5:exit
Enter your choice:2
The popped element is: 30
1:push 2:pop 3:display 4:peek 5:exit
Enter your choice:4
Top item is 20
1:push 2:pop 3:display 4:peek 5:exit
Enter your choice:
Assistant
Warnings
May be ignored if you are happy with your program.
queue.py
Possibly bad file name
bracketmatch.py
Line 1 : Redefining name 'expr' from outer scope (line 29)
Was it helpful or confusing?
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Type here to search
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ENG IN
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1/23/2025

11. Implement Bracket Matching using stack.

Algorithm:

```
def  isBalanced(expr):
    stack = []
    for char in expr:
        if char in ["(", "{", "["]:
            stack.append(char)
        else:
            # Here we check if the current character is not opening
            # bracket, then it must be closing.
            # So stack cannot be empty at this point.
            if not stack:
                return False
            curr_char = stack.pop()
            if curr_char == '(':
                if char != ")":
                    return False
            if curr_char == '{':
                if char != "}":
                    return False
            if curr_char == '[':
                if char != "]":
                    return False

    # Here we check empty stack
    if stack:
        return False
    return True
```

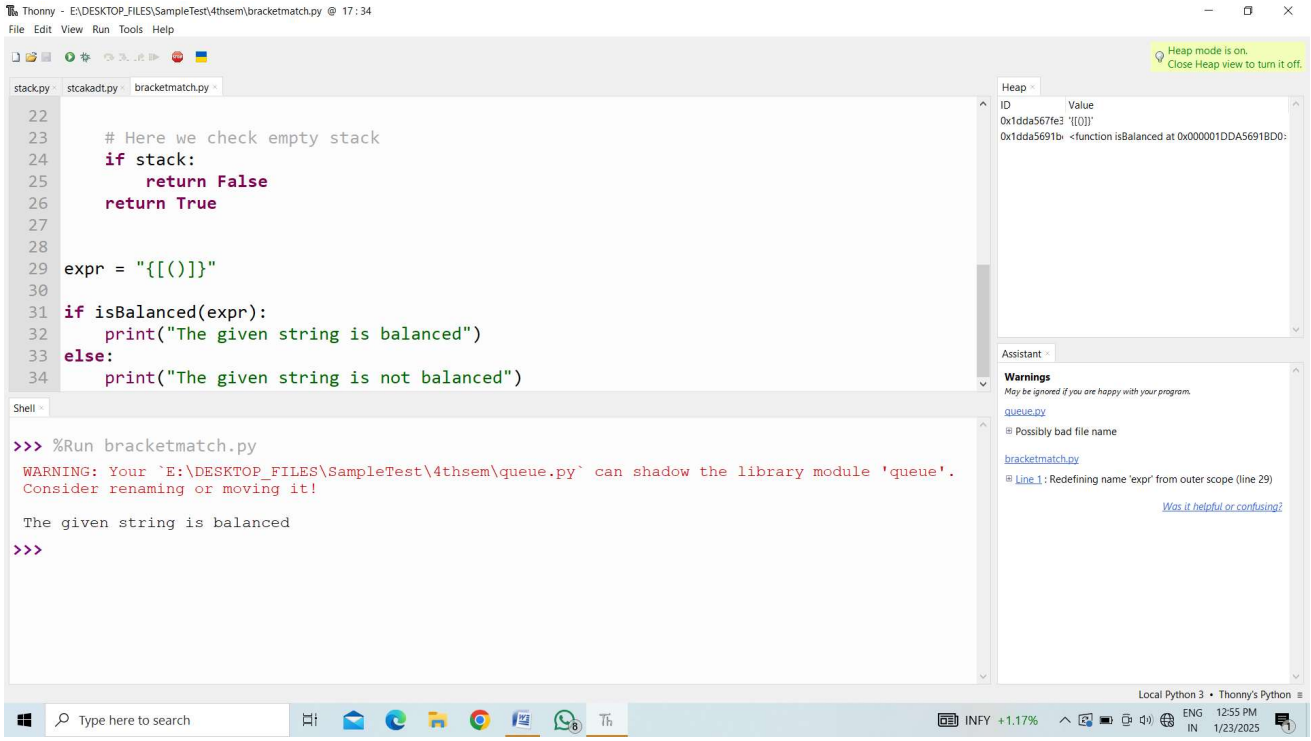
```
def  isBalanced(expr):
   stack = []
   for char in expr:
      if char in ["(", "{", "["]:
         stack.append(char)
      else:
         # Here we check if the current character is not opening
         # bracket, then it must be closing.
         # So stack cannot be empty at this point.
         if not stack:
            return False
         curr_char = stack.pop()
         if curr_char == '(':
            if char != ")":
               return False
         if curr_char == '{':
            if char != "}":
               return False
         if curr_char == '[':
            if char != "]":
               return False

   # Here we check empty stack
   if stack:
      return False
   return True

  expr = "{[()]}"

if  isBalanced(expr):
     print("The given String is balanced")
else:
     print("The given String is not balanced")
```

Output:


Thonny - E:\DESKTOP_FILES\SampleTest\4thsem\bracketmatch.py @ 17 : 34
File Edit View Run Tools Help
Heap mode is on.
Close Heap view to turn it off.
stack.py
stcakadt.py
bracketmatch.py
# Here we check empty stack
if stack:
return False
return True
expr = "{[()]}"
if isBalanced(expr):
print("The given string is balanced")
else:
print("The given string is not balanced")
Heap
ID
Value
'{[()]}'
<function isBalanced at 0x000001DDA5691BD0>
Shell
>>> %Run bracketmatch.py
WARNING: Your `E:\DESKTOP_FILES\SampleTest\4thsem\queue.py` can shadow the library module 'queue'.
Consider renaming or moving it!
The given string is balanced
>>>
Assistant
Warnings
May be ignored if you are happy with your program.
queue.py
Possibly bad file name
bracketmatch.py
Line 1 : Redefining name 'expr' from outer scope (line 29)
Was it helpful or confusing?
Local Python 3 • Thonny's Python
Type here to search
INFY +1.17%
ENG
IN
12:55 PM
1/23/2025

12. Program to demonstrate recursive operations-Factorial

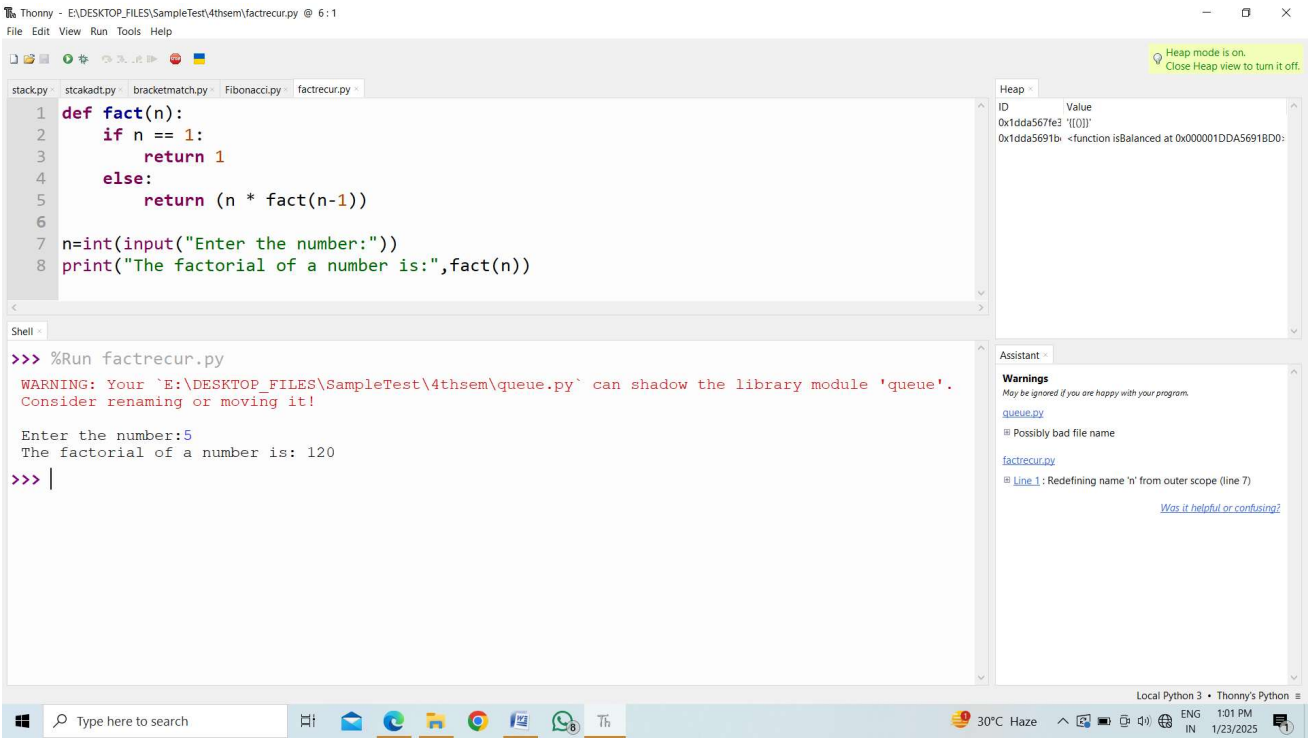
Algorithm:

```
def  fact(n):
     if n == 1:
         return 1
    else:
        return (n * fact(n-1))
```

```
def fact(n):
   if n == 1:
      return 1
   else:
      return (n * fact(n-1))

n=int(input("Enter the number:"))
print("The factorial of a number is:",fact(n))
```

Output:



12a. Program to demonstrate recursive operations-Fibonacci

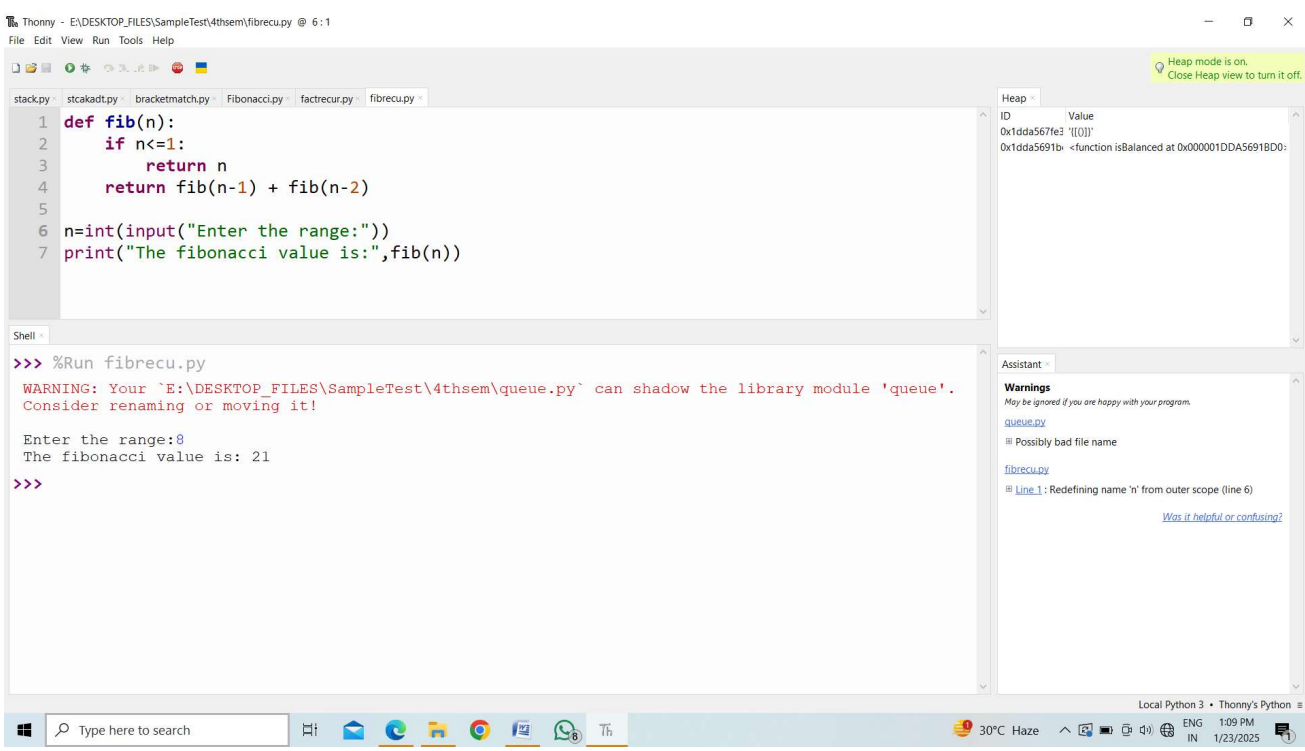
Algorithm:

```
def fib(n):
    if n<=1:
        return n
    return fib(n-1) + fib(n-2)
```

```
def fib(n):
    if n<=1:
        return n
    return fib(n-1) + fib(n-2)

n=int(input("Enter the range:"))
print("The fibonacci value is:",fib(n))
```

Output:



13. Implement Queue Data Structure.